

Multi-Paradigm Approaches Toward General-Purpose Embodied Intelligence: Panoramic Overview and Evolutionary Trajectory

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Abstract—Vision-Language-Action Models (VLAs) have emerged as an important approach to general-purpose robotic control by connecting visual perception, language instructions, and action generation. In recent years, VLA research has evolved from early language-conditioned policies toward larger generalist models and more predictive forms of embodied decision-making. However, existing surveys mainly examine architectures, datasets, or individual learning paradigms, with limited attention to the interplay among action generation, deployment, generalization, and predictive modeling. This survey reviews this development and organizes representative methods according to their action-generation mechanisms. We summarize major VLA architectures and commonly used action decoders, and discuss Video-Action Models and World Action Models as emerging extensions that incorporate temporal dynamics and predictive world modeling into robot control. We then review representative benchmark results together with recent progress toward efficient deployment and robust generalization. Finally, we summarize the main challenges that remain for real-world use and discuss future directions for more robust and adaptive robotic policies.

Index Terms—Vision-Language-Action, World Action Models, Video-Action Models, Robot Learning, Embodied Intelligence.

I. Introduction

Recent advances in artificial intelligence have increasingly extended beyond static perception and semantic understanding toward embodied systems that interact with the physical world. Embodied agents must not only understand natural language instructions and perceive complex environments, but also generate executable actions and continuously adjust these actions based on environmental feedback. Vision-Language-Action Models (VLAs) provide an important technical path toward general-purpose robotic policies by learning a unified mapping from visual observations and language instructions to robot actions.

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The project repository for this article is available at <https://github.com/RuleShi/VLA-Embodied-Intelligence-Survey>.

The accompanying benchmark leaderboard is available at <https://embodiedbench.vercel.app/en>.

RT-1 [1] validated the scalability of Transformers for large-scale, multi-task robotic control; RT-2 [2] transferred knowledge from vision-language pretraining to action generation; RT-X models [3] further explored joint training across tasks and robots; OpenVLA [4] promoted the open-sourcing of models and training pipelines; and π_0 [5] adopted a flow-matching mechanism to generate continuous action chunks. These approaches each have advantages in inference efficiency, semantic modeling, or continuous control, resulting in different trade-offs among control precision, generation speed, and model expressiveness.

Although VLAs have made significant progress in multi-task control and semantic generalization, their real-world deployment remains constrained by computational overhead, data dependency, environmental variation, and limitations in long-horizon execution. In terms of model efficiency, visual token pruning, quantization, knowledge distillation, and lightweight backbones can reduce computational load and storage cost, but model-level compression does not necessarily translate into high-frequency, low-latency, and stable closed-loop control in real robotic systems. In terms of scene generalization, data resources such as Open X-Embodiment [3] and DROID [6] have expanded the coverage of tasks, objects, and robot embodiments, while benchmarks such as LIBERO [7], RoboCasa365 [8], and RoboTwin 2.0 [9] have introduced richer task configurations and environmental variation. However, high success rates in standard or simple scenarios do not imply that a model can reliably handle variations in camera viewpoint, object layout, lighting, background, or robot initial state; even minor perturbations outside the training distribution can still cause a marked performance drop. Meanwhile, recent studies have explored reinforcement learning, memory, hierarchical planning, and world modeling to move VLAs beyond reactive action prediction toward more adaptive and temporally coherent decision-making. Despite this progress, achieving reliable long-horizon behavior in real-world settings remains challenging, particularly under limited interaction data and distribution shifts [10]–[13]. Therefore, the key to moving VLAs toward real-world application does not lie solely in pursuing smaller models or broader data coverage, but rather in simultaneously balancing deployment efficiency, robust generalization, and long-term decision consistency.

Beyond improving the efficiency and generalization of action prediction itself, recent research has begun to incorporate video dynamics and future world states into robotic policy learning. Early world-model research showed that

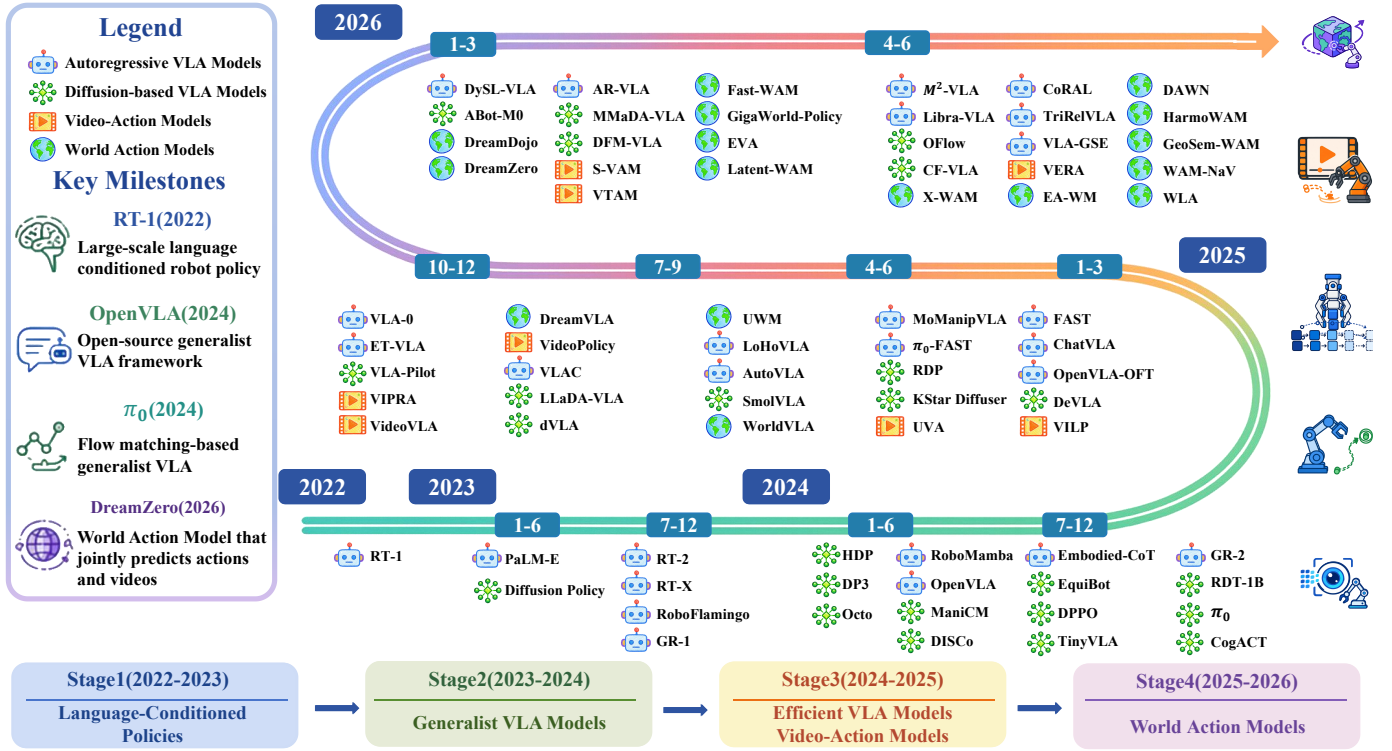


Fig. 1. Chronological roadmap of representative embodied robot policies, VLAs, VAMs, and WAMs from 2022 to 2026. The evolution progresses from language-conditioned policies to generalist and efficient VLAs, and subsequently to VAMs and WAMs. Different colors denote the major action-generation and predictive modeling paradigms. Years in the roadmap denote the first public release of each method.

an agent can learn a latent dynamic representation of the environment to predict subsequent states and perform behavioral rollouts within an internal model [14]. Building on this, Video-Action Models (VAMs) leverage temporal changes, object motion, and interaction information within videos to assist action generation—for example, UVA [15] establishes a shared video-action representation by jointly learning future observations and actions; World Action Models (WAMs) further emphasize the joint modeling of future world states and action sequences, enabling policies to account for the environmental changes that actions may induce [16]. VAMs and WAMs are not simple replacements for existing VLAs, but rather emerging directions that extend VLAs toward modeling temporal dynamics, action consequences, and future states. Existing surveys of robotic foundation models mainly focus on model types, learning paradigms, tasks, and applications [10], [17], [18]; VLA surveys focus on model components, control strategies, and high-level planning [11], [12]; data-centric surveys focus on datasets, benchmarks, and data engines [13]; and WAM surveys mainly discuss how world prediction and action generation are fused [16].

However, existing studies rarely examine, from a unified perspective, how action-generation architectures, deployment-oriented design, and predictive modeling jointly shape the capabilities and real-world applicability of VLAs. To address this gap, we organize the literature around action generation as the architectural backbone, deployment efficiency and scene generalization as practical

considerations, and VAMs/WAMs as emerging extensions toward temporal and future-state modeling.

The main contributions of this paper are as follows:

- 1) **Unified Taxonomy of VLA Architectures.** We construct a systematic architectural taxonomy of VLAs, reviewing their core components in terms of vision-language encoding, LLM backbones, and action decoding, with particular emphasis on summarizing the two major action-generation paradigms of autoregression and diffusion.
- 2) **Deployment Efficiency and Robust Generalization.** From a deployment and generalization perspective, we review the key methods enabling VLAs to move toward real-world application, systematically summarizing frontier progress in lightweight deployment and cross-scene generalization, and analyzing their value and limitations.
- 3) **Boundary and Relationship Clarification of VAMs/WAMs and VLAs.** We clarify the technical boundary between VAMs and WAMs and their relationship to VLAs, pointing out that VAMs focus more on the coupling between video temporal dynamics and action sequences, while WAMs further emphasize the joint prediction of future world states and actions, representing an important frontier direction for VLAs moving toward predictive embodied decision-making.
- 4) **Unified Benchmark Overview and Online Leaderboard.** We provide a unified overview of

widely used VLA benchmarks and develop an on-line leaderboard that aggregates publicly reported results across multiple benchmarks for convenient model comparison and progress tracking.

- 5) **Open Challenges and Future Embodied Intelligence.** Building on the preceding analysis, we reveal the main bottlenecks VLAs face on the path toward real-world deployment and further outline their future development trajectory.

The remainder of this paper is organized as follows: Section II introduces the core architecture of VLAs, their action-generation mechanisms, VAMs and WAMs, and summarizes related datasets and evaluation benchmarks; Section III discusses model efficiency, scene generalization, and other capability-enhancement methods; Section IV summarizes the main current challenges; Section V summarizes representative applications; Section VI looks ahead to future research directions; and Section VII concludes the paper.

II. VLA Architecture: Core Components and Their Evolution

A. Overview of VLAs

As depicted in Fig. 2, VLAs aim to unify visual perception, language understanding, and robotic action generation within a single decision-making framework, thereby enabling embodied control driven by natural language instructions. A typical VLA pipeline [1], [4], [5], [19], [20] can be summarized as encoding of vision and language inputs, an LLM backbone network, and an action-generation module.

- 1) *Vision Encoder:* The vision encoder is the perceptual front-end of a VLA, responsible for converting observed images into visual feature embeddings that can be used for cross-modal reasoning and action decision-making. Existing VLA systems typically rely on vision foundation models pretrained on large-scale image or image-text paired data, such as ViT [21], CLIP [22], SigLIP [23], and DINOv2 [24], to strengthen the model’s ability to represent object semantics, spatial relationships, and scene context. Among these, ViT demonstrates the scalability of modeling sequences of image patches for large-scale visual pretraining; CLIP and SigLIP strengthen the association between visual representations and linguistic semantics through image-text alignment learning; and DINOv2 shows that self-supervised visual pretraining can produce general-purpose features with strong transferability. Overall, these vision foundation models provide important support for open-scene perception and cross-task transfer in VLAs.

- 2) *Language Instruction:* The language instruction is the semantic interface connecting human task intent with robotic action generation in a VLA, typically used to describe manipulation targets, goal states, spatial relationships, execution order, and behavioral constraints. After being processed by a tokenizer or text encoder, the instruction is converted into language tokens, which together

with the visual observation and the robot’s proprioceptive state form the conditioning input to the policy model.

- 3) *LLM Backbone:* In many modern VLA architectures, the LLM backbone serves as the core module for contextual and semantic modeling. It processes language instructions together with visual representations, robot states, and historical context, and uses Transformer-based sequence modeling to capture task semantics and interaction dependencies. The resulting task-conditioned representations are then provided to the action decoder for robot action generation.

Large-scale autoregressive pretraining provides a strong foundation for such backbones. Pretrained models such as GPT-2 [25], Llama 2 [26], Qwen [27], and Gemma [28] offer transferable language representations and contextual modeling capabilities, allowing VLA systems to better interpret task instructions and organize multimodal information. Related embodied models further illustrate how Transformer-based backbones can be adapted to robot control: Gato [29] represents text, images, observations, and actions within a unified sequence model, while RoboFlamingo [30] combines a pretrained vision-language backbone with a dedicated policy head for sequential action generation.

- 4) *Action Decoder:* The action decoding head is a key component of VLAs that bridges multimodal semantic understanding and robotic control. It maps the high-dimensional representations generated by vision-language encoders and multimodal backbones into executable robot actions, such as joint angles, end-effector poses, and gripper states. Since visual and linguistic representations are typically abstract and semantically rich, whereas robotic actions reside in a continuous, low-dimensional, and physically constrained control space, the action decoding head serves as a critical interface for translating high-level semantic representations into low-level physical actions.

In terms of how actions are generated, existing action decoders can broadly be divided into three categories: MLP-based, autoregressive, and diffusion-based. MLP decoders are structurally simple and computationally efficient, typically used to directly regress continuous actions; autoregressive decoders represent actions as tokens or sequence units and generate actions through sequential prediction; diffusion-based decoders model continuous action trajectories through a conditional generative process. Compared with MLP decoders, autoregressive and diffusion-based decoders better reflect the main development trends in large-scale action modeling for current VLAs, so the following analysis focuses primarily on these two categories.

- a) *Autoregressive Action Decoders:* Autoregressive action decoders formulate robot action generation as a conditional sequence-prediction problem. The core idea is to discretize or serialize continuous control signals into action tokens and generate them sequentially in causal order, conditioned on task-relevant multimodal representations and previously generated actions. By representing actions within a sequential prediction framework, autoregressive

decoders can leverage the contextual modeling and semantic reasoning capabilities provided by LLM backbones while adapting them to robot control.

RT-H [31] builds a hierarchical decision structure through languaged actions and autoregressively generates discrete action tokens in causal order; FAST [32] uses discrete cosine transform and BPE to compress continuous action chunks into shorter token sequences, then predicts subsequent actions step by step based on already-generated tokens, thereby reducing the sequence-modeling overhead of high-frequency control; AR-VLA [33] introduces an independent autoregressive action expert together with a long-term memory mechanism, allowing subsequent actions to continuously depend on historical action context and periodically updated vision-language information, thereby achieving continuous action-stream generation.

b) Diffusion-Based Action Decoders: Such methods generate continuous action sequences through an iterative denoising process, typically starting from a noise distribution or an action prior and conditioning on task-relevant multimodal representations. Unlike autoregressive decoders that often rely on discrete or serialized action representations, diffusion-based decoders operate in continuous action space and are well suited to modeling complex and multimodal action distributions, where multiple valid action sequences may exist under the same task condition. It should be noted that flow matching also predicts actions through a continuous generative process, but its training objective and generative mechanism are not entirely identical to those of diffusion models; this paper discusses flow matching as a continuous generative action route closely related to diffusion methods.

Representative studies have improved diffusion-based action generation along several dimensions, including architectural design, model scaling, and sampling efficiency. CogACT [34] decouples the vision-language cognitive module from the diffusion action expert to strengthen the functional division between high-level semantic understanding and low-level continuous control; RDT-1B [35] further scales up the diffusion action Transformer and uses a unified action representation to handle heterogeneous robots and bimanual coordination tasks. To address the inference latency caused by multi-step denoising, ManiCM [36] uses consistency distillation to compress action generation into a single or a few inference steps, while CF-VLA [37] adopts a coarse-to-fine flow-generation strategy to improve action initialization and reduce the number of iterations.

In summary, VLA action decoding has not yet converged on a unified paradigm; instead, three routes—autoregressive, diffusion-based, and MLP-based—have formed around trade-offs among semantic generalization, control precision, and inference efficiency. Autoregressive decoders make it convenient to reuse the knowledge and reasoning capability of large language models, but are constrained by action discretization error and token-by-token generation latency; diffusion-based decoders and

flow matching are better at modeling continuous, multimodal action distributions and can generate smooth, fine-grained trajectories, but incur higher inference overhead; MLP decoders are structurally simple, highly real-time, and suitable for lightweight deployment, but have limited expressive power for complex action distributions. Therefore, the three approaches are not mutually substitutable, but rather trade-offs suited to different task requirements. Future action decoders may move further toward hybrid and hierarchical designs to jointly balance semantic reasoning, continuous control, and deployment efficiency.

B. VAMs: Robot Manipulation Prediction Models that Jointly Model Video Dynamics and Action Sequences

In recent years, advances in large-scale video generation models and robotic foundation models have made video dynamics increasingly relevant to embodied policy learning. VAMs model the temporal evolution of task execution from continuous video and use the resulting predictive information to guide robot actions. By capturing how visual states and object interactions evolve over time, these models provide temporal priors that can improve the continuity and physical consistency of action generation.

Based on the primary path by which video dynamics participate in action generation, existing VAM approaches can be broadly categorized into two types: coupled video-action generation and cascaded video-planning-to-action conversion. This categorization is not strictly mutually exclusive; for methods that combine both joint modeling and cascaded decoding characteristics, this paper classifies them according to their dominant inference structure.

The first category typically couples video prediction and action generation through shared latent representations or parallel generative branches. UVA [15] learns a joint video-action latent representation with decoupled decoding, allowing full video generation to be bypassed during inference for more efficient action prediction. VideoVLA [38] adapts pretrained video-generation models to robotic manipulation and jointly predicts future visual outcomes and action sequences conditioned on language and images. CoVAR [39] employs parallel video- and action-diffusion branches with Bridge Attention to enable cross-modal interaction and joint video-action generation. DiT4DiT [40] extracts intermediate features from the video-denoising process and uses them as conditions for action prediction, jointly training the video and action modules within a cascaded framework.

The second category follows a video-planning-to-action pipeline, in which future visual states or latent trajectories are first predicted and then translated into robot actions through a policy decoder or an inverse-dynamics model. VILP [41] generates temporally consistent predictive videos in latent space and uses them to support downstream policy learning. ViPRA [42] learns motion-centric latent actions from action-unlabeled videos and maps them to robot-specific continuous action sequences through a flow-matching decoder. VERA [43]

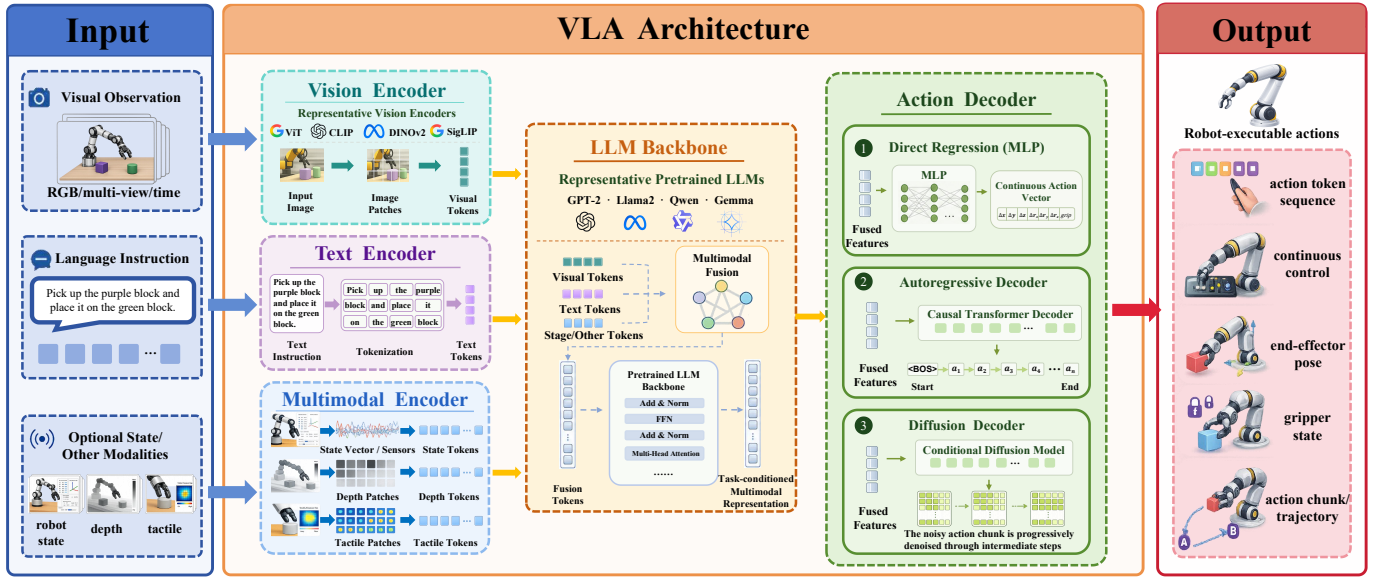


Fig. 2. Architectural overview of VLAs. Encoding: Heterogeneous inputs are encoded and aligned through multimodal fusion. Multimodal reasoning: An LLM backbone processes the fused representations to produce task-conditioned features for decision-making. Action decoding: The resulting representations are mapped into executable robot actions through direct regression, autoregressive prediction, or diffusion/flow matching.

keeps the video planner separate from an embodiment-specific inverse-dynamics model, allowing the same visual planner to be paired with different robot embodiments. This modular design separates visual prediction from robot-specific action translation, enabling the two components to be trained or adapted with different data sources.

Beyond architectural design, another important direction is to exploit large-scale video data for robot policy learning. Pretrained video models can provide temporal and interaction priors learned from internet-scale or human manipulation videos, reducing the reliance on densely action-labeled robot demonstrations. These learned priors can provide useful cues about object motion and interaction dynamics that may be difficult to capture from limited robot data alone. *mimic-video* [44] leverages latent representations from pretrained video models to support action learning, while *Mitty* [45] explores transferable manipulation knowledge from action-unlabeled human videos.

Despite this progress, VAMs still face practical limitations, particularly the computational cost of video models and the mismatch that may arise between predicted visual dynamics and executable robot actions. In particular, visually plausible future predictions do not necessarily correspond to physically feasible actions for a specific robot embodiment. Further progress will depend on more efficient dynamics modeling and tighter integration between video prediction and closed-loop control.

C. WAMs: Predictive Policy Learning Fusing World-State Prediction with Action Generation

With the development of video-generation models, world models, and robotic foundation models, WAM has gradually become an important direction for embodied policy learning. Compared with VLAs, which primarily

optimize action generation conditioned on observations, WAM further incorporates future world-state prediction into the action-decision process, enabling models to generate, evaluate, or optimize actions based on anticipated environmental changes. Its core is not merely to generate future images or video, but to establish a connection among world dynamics, action consequences, and control policy, thereby enhancing the robot's ability to model physical change and long-term task processes [16].

WAMs and VAMs overlap in terms of future visual prediction, but the two differ in emphasis. VAMs primarily use the temporal features learned by video models to enhance action prediction, while WAMs are more concerned with the functional coupling between future states and action decisions. Future states can be represented not only as images and video, but also as latent features, object states, 3D geometry, or multimodal physical representations. Accordingly, WAMs are better characterized by whether predictive world modeling is explicitly integrated into the action-generation process, rather than by whether they generate future videos.

In terms of the role future states play in the policy, existing WAM approaches first manifest as world-action modeling driven by explicit prediction. *DreamZero* [46] uses a pretrained video-diffusion model to jointly predict future visual states and action sequences, allowing video dynamic priors to participate in robot control; *World-VLA* [47] organizes future images and actions within a unified autoregressive sequence, strengthening the association between state evolution and action prediction. This class of methods can provide relatively intuitive future representations, but explicitly generating high-dimensional visual states also increases inference cost and may propagate world-prediction errors into the action output.

To reduce the overhead brought by future generation, recent research has begun to convert world prediction into latent dynamic representations, training-time supervision, or optional inference branches. Fast-WAM [48] retains the future video-modeling objective during training but removes the full video-generation branch at inference time, directly conditioning action prediction on latent world features; GigaWorld-Policy [49] further adopts an action-centered design, using world prediction as auxiliary supervision for training the action policy. HarmoWAM [50] uses an adaptive mechanism to coordinate a world model with strong generalization capability with an action expert oriented toward fine control, while SANTS [51] dynamically adjusts the world-generation and action-execution process based on the current state. Related research shows that the value of WAM is not limited to generating visualizable futures, but is also reflected in strengthening the model’s consistency in modeling environmental dynamics and actions through prediction objectives.

Recent WAM studies have increasingly explored structured world representations and tighter interaction between world prediction and action generation, moving beyond purely holistic visual prediction. OA-WAM [52] adopts an object-addressable representation, separately modeling the robot, target objects, and their time-varying states, enhancing target localization and interaction robustness in complex scenes; X-WAM [53] fuses multi-view RGB-D information, extending world prediction to a 4D representation that includes spatial geometric change; GeoSem-WAM [54] further jointly leverages geometric structure and semantic cues to improve action-conditioned world modeling capability. DAWN [55] enables world prediction and action generation to recursively refine each other during inference, extending WAM to interactive autonomous-driving planning; WorldVLN [56] uses an autoregressive video backbone to predict short-horizon latent world-state transitions and decodes them into executable waypoints, continuously updating its prediction context under real observational feedback, demonstrating a closed-loop implementation of WAM in aerial vision-language navigation.

Overall, WAMs extends VLAs from observation-conditioned action generation to predictive decision-making informed by anticipated world evolution. Despite these advances, WAMs remain sensitive to errors in predicted world states, which may propagate into action decisions, while explicit world generation can introduce substantial inference overhead. Further progress will require more reliable predictive representations and more efficient integration of world modeling with closed-loop action generation.

D. Datasets and Benchmarks

Existing VLA datasets and benchmarks have gradually expanded from multi-task manipulation and skill transfer toward language-conditioned control, long-horizon task execution, and robustness-oriented generalization. This

evolution reflects a broader shift in evaluation from measuring whether a policy can complete individual manipulation tasks to assessing whether it can interpret instructions, compose behaviors over time, and remain reliable under variations in tasks and environments. As illustrated in Fig. 3, representative benchmarks, such as Meta-World [78], LIBERO [7], CALVIN [80], LIBERO-Plus [79], and RoboTwin 2.0 [9], characterize the evolution of robotic learning evaluation by targeting different levels of task complexity, generalization, and real-world applicability.

1) *Meta-World*: Meta-World is a simulated robotic manipulation benchmark containing 50 distinct manipulation tasks with shared object and interaction structures. It is designed to evaluate the capabilities of reinforcement learning and imitation learning algorithms in multi-task learning, task transfer, and generalization to unseen tasks. Meta-World defines several evaluation settings, including the ML series for meta-learning scenarios and the MT series for multi-task learning scenarios. The ML settings evaluate adaptation to new task distributions through limited interaction experience, whereas the MT settings evaluate joint learning across multiple manipulation tasks. Due to its standardized simulation environments and diverse task collection, Meta-World has been widely adopted for evaluating robotic manipulation policies [78].

2) *LIBERO*: LIBERO is a lifelong robot learning benchmark designed to study knowledge transfer in robotic manipulation. It focuses on transferring different types of knowledge, including task semantics, object configurations, and manipulation procedures, across sequential learning settings. LIBERO provides a procedural task-generation pipeline and contains 130 manipulation tasks organized into four suites: LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-100. LIBERO-Spatial evaluates spatial knowledge transfer, LIBERO-Object focuses on object-level knowledge transfer, LIBERO-Goal studies goal-related knowledge transfer, and LIBERO-100 contains a larger collection of diverse manipulation tasks for lifelong learning evaluation. In addition, LIBERO provides high-quality human teleoperation demonstrations, making it suitable for imitation learning, lifelong robot learning, and vision-language-action model evaluation [7].

3) *CALVIN*: CALVIN is a simulated benchmark for language-conditioned long-horizon robotic manipulation. It contains 34 composable manipulation tasks involving interactions such as block manipulation, drawer opening, sliding-door operation, and light-switch control. The benchmark evaluates long-horizon task execution by composing basic tasks into sequential instruction chains, requiring robots to complete multiple subtasks according to natural-language commands. CALVIN provides multimodal observations and supports different action representations, including absolute Cartesian poses, relative Cartesian displacements, and joint actions. Therefore, it has been widely used to evaluate language-conditioned policy learning, compositional generalization, and long-

TABLE I
REPRESENTATIVE EMBODIED FOUNDATION MODELS WITH DIFFERENT ACTION GENERATION AND PREDICTION PARADIGMS

Model	Year	Key innovation
Autoregressive VLA		
RT-2 [2]	2023	Tokenizes robot actions and transfers web-scale vision-language knowledge to robotic control through VLM co-fine-tuning.
RT-H [31]	2024	Builds a hierarchy from languaged actions and autoregressively generates discrete action tokens.
OpenVLA [4]	2025	Discretizes continuous actions into tokens for autoregressive robot control with a pretrained VLM.
ECot [57]	2025	Autoregressively generates a chain of task, perceptual, and motion reasoning before acting.
π_0 -FAST [32]	2025	Tokenizes action chunks with DCT and BPE for efficient autoregressive action prediction.
ChatVLA [58]	2025	Unifies multimodal understanding and robot control through staged alignment and MoE.
ET-VLA [59]	2025	Adapts VLAs to multi-robot embodiments via synthetic continued pretraining and Embodied Graph-of-Thought reasoning.
AutoVLA [60]	2025	Unifies trajectory generation with fast/slow reasoning and uses GRPO-based reinforcement fine-tuning for adaptive reasoning.
LoHoVLA [61]	2025	Uniformly generates subtask and action tokens, executing long-horizon tasks via hierarchical closed-loop control.
AR-VLA [33]	2026	Introduces a true autoregressive action expert with a hybrid KV cache for continuous action-stream prediction.
Diffusion-based VLA		
π_0 [5]	2025	Combines a VLM with a flow-matching action expert to generate continuous action chunks.
TinyVLA [63]	2025	Combines a lightweight VLM with a diffusion head, requiring no robot-data pretraining.
DexVLA [64]	2025	Proposes a 1B-parameter diffusion action expert trained with a three-stage embodied curriculum.
SmolVLA [65]	2025	A lightweight VLM with a flow-matching action expert supports asynchronous, low-cost deployment.
DreamVLA [66]	2025	Predicts dynamics, depth, and semantic knowledge to guide diffusion-based action generation.
LLaDA-VLA [67]	2025	Combines a diffusion VLM with action-specific classification and hierarchical decoding.
dVLA [68]	2025	Unifies a discrete diffusion objective to jointly generate multimodal chain-of-thought and actions.
CF-VLA [37]	2026	Replaces pure-noise initialization with an action prior via two-stage "coarse-to-fine" generation.
VLA-Pilot [69]	2026	Optimizes actions at inference time via evolutionary diffusion, enabling fine-tuning-free deployment.
ProgressVLA [70]	2026	Guides diffusion sampling with progress estimation, optimizing execution and termination of long-horizon tasks.
VAM		
UVA [15]	2025	Jointly learns a video-action latent representation and enables fast action inference via a decoupled diffusion head.
VILP [41]	2025	Generates multi-view-consistent video plans in latent space, mapped to actions via a closed-loop low-level policy.
ViPRA [42]	2025	Jointly predicts future vision and latent motion actions, generating continuous action chunks via a flow-matching decoder.
mimic-video [44]	2025	Conditions a flow-matching action decoder on pretrained video latent representations to improve data efficiency.
VideoVLA [38]	2025	Jointly predicts future video and actions based on a pretrained video generator, enhancing manipulation generalization.
CoVAR [39]	2025	Sets up parallel video and action diffusion branches, cogenerated video-action pairs via Bridge Attention.
S-VAM [71]	2026	Self-distills a multi-step video prior into single-step geometric-semantic foresight for real-time action prediction.
VTAM [72]	2026	Jointly predicts visual-tactile dynamics and enhances contact control with virtual force targets.
DiT4DiT [40]	2026	Jointly trains video and action DiTs, conditioning action generation on video-denoising features.
VERA [43]	2026	Achieves closed-loop cross-embodiment control via a decoupled video planner and a J-IDM.
WAM		
UWM [73]	2025	A unified diffusion framework that flexibly achieves policy, dynamics, and video generation via decoupled timesteps.
DreamDojo [74]	2026	Large-scale pretraining on human videos, introducing "continuous latent actions" to address the use of unlabeled data.
Fast-WAM [48]	2026	Eliminates test-time imagination, showing that training-time video-model synergy drives improved performance.
X-WAM [53]	2026	Combines unified 4D modeling with asynchronous denoising, balancing high-fidelity synthesis with real-time execution.
HarmoWAM [50]	2026	Uses adaptive gating to coordinate predictive and reactive experts for generalizable transit and precise manipulation.
OA-WAM [52]	2026	Introduces object-addressable slots, improving the model's interaction robustness in complex scenes.
CKT-WAM [75]	2026	Enables efficient knowledge transfer between heterogeneous models via a compressed interface in text-embedding space.
Latent-WAM [76]	2026	Combines geometric distillation with a causal Transformer for efficient latent-space planning.
τ_0 -WM [77]	2026	Unifies video-action policy learning and action-conditioned video simulation within a shared world-model backbone.
DreamZero [46]	2026	Jointly predicts future video and actions to learn physical dynamics and enable cross-embodiment transfer.

horizon manipulation capabilities [80].

4) *LIBERO-Plus*: LIBERO-Plus is a robustness evaluation framework built upon the LIBERO benchmark to analyze the generalization capability of vision-language-action models under environmental perturbations. It introduces diverse perturbation factors, including object layout, camera viewpoint, robot initial state, language instructions, lighting conditions, background texture, and sensor noise. By evaluating models under these controlled distribution shifts, LIBERO-Plus aims to measure the robustness of robotic policies beyond standard benchmark settings [79].

5) *RoboTwin 2.0*: RoboTwin 2.0 is a data-generation and evaluation platform designed for bimanual robotic manipulation. It provides a collection of manipulation tasks covering various interaction scenarios, including object pick-and-place, bimanual handover, stacking, sorting,

articulated-object manipulation, tool use, and fine-grained interaction. The benchmark supports multiple robot platforms and improves data diversity through variations in scene configurations, lighting conditions, backgrounds, tabletop heights, and language instructions. RoboTwin 2.0 evaluates policies under an Easy clean setting and a Hard domain-randomized setting, allowing robustness to environmental variation to be measured under a unified protocol [9].

In addition to the above benchmarks, RLBench [81], RoboCasa365 [8], RoboCasa-GR1 [82], BEHAVIOR-1K [83], and SimplerEnv [84] provide complementary resources for VLA training and evaluation. RLBench provides a collection of simulated manipulation tasks with multimodal observations and demonstration data, and has been widely used in reinforcement learning, imitation learning, and multi-task policy evaluation. Built on the

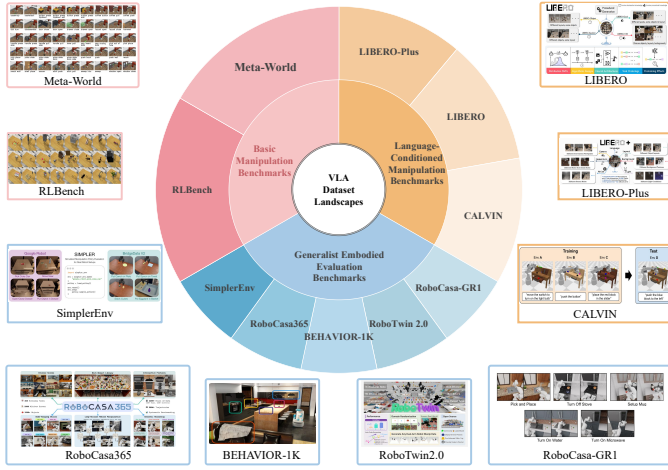


Fig. 3. Landscape of datasets and evaluation benchmarks for generalist VLAs. Representative benchmarks are broadly organized according to their primary evaluation focus into basic manipulation, language-conditioned manipulation, and generalist embodied evaluation. Outer callouts illustrate representative simulation setups and task environments.

RoboCasa platform, RoboCasa365 provides a large-scale simulation benchmark with 365 everyday household tasks across diverse kitchen environments for training and evaluating generalist robot policies. RoboCasa GR-1 Tabletop Tasks provides 24 simulated tabletop manipulation tasks for the GR-1 humanoid robot and supports closed-loop evaluation of generalist robot policies in household manipulation settings. BEHAVIOR-1K provides 1,000 realistic household activities and emphasizes long-horizon embodied interaction in complex, human-centered environments. SimplerEnv provides simulation environments for evaluating real-world robot manipulation policies and studies how well simulated evaluation reflects their real-world behavior. Overall, these benchmarks complement the core VLA evaluation landscape by extending evaluation to more diverse and practically relevant settings.

III. Deployment and Generalization

A. Model Efficiency

Despite these advances, scaling VLAs to real-world deployment remains challenging. As model parameter scale grows and multimodal fusion becomes increasingly complex, VLAs face high computational overhead and low inference efficiency. RT-2 notes that its computational cost limits real-time inference in high-frequency control scenarios and suggests optimization through quantization and distillation. OpenVLA further observes that many existing VLAs remain closed-source and lack deployment practices for new robots, environments, and tasks, especially on consumer-grade hardware [4]. Subsequent studies have also highlighted the challenges of training and inference costs and limited real-time performance [63], [85], [86]. This motivates recent efforts in visual token pruning, quantization, and distillation to reduce computational overhead while preserving performance and generalization.

1) *Visual Token Pruning*: Visual token pruning reduces self-attention overhead by removing redundant visual tokens from Transformers. Unlike pruning methods for general VLAs that mainly rely on visual saliency, token selection in VLAs must also consider action relevance, temporal dependency, and geometric structure to avoid discarding information critical to manipulation. Accordingly, recent studies have gradually shifted from perception-driven pruning toward action- and task-aware strategies.

Training-free methods can be directly integrated into existing VLAs without additional optimization. VLA-IAP [87] preserves key manipulation structures using geometric priors and dynamically adjusts pruning strength. DTP [88] removes visual tokens from task-irrelevant regions to refine visual attention. ADP [89] adaptively adjusts the token retention ratio according to action trajectories. SpecPrune-VLA [90] performs dynamic pruning based on action temporal information and layer-wise token importance. These methods mainly exploit action, motion, or geometric priors to improve token selection while reducing computation without retraining.

Training-based methods learn task-relevant token selection through fine-tuning or end-to-end optimization. DeepVision-VLA [91] removes irrelevant tokens using an action-guided visual pruning module. BFA++ [92] reduces spatial and multi-view redundancy through intra-view and inter-view predictors. ETA-VLA [93] compresses cross-frame tokens by combining text relevance with temporal consistency. LightVLA [94] adopts differentiable pruning to learn token retention strategies. CogVLA [95] removes instruction-irrelevant information through progressive sparsification, while VLA-Pruner [96] evaluates and compresses redundant tokens according to action relevance. Compared with training-free methods, these approaches can learn finer-grained task-specific token importance through optimization.

System-level co-optimization further combines token pruning with other acceleration techniques. SP-VLA [97] integrates token pruning with dynamic model scheduling. AC²-VLA [98] jointly performs feature reuse, token pruning, and selective component execution. SQAP-VLA [99] combines quantization and pruning for joint acceleration. These methods extend beyond token compression toward multidimensional computational optimization, further improving end-to-end inference efficiency.

2) *Quantization*: Quantization reduces memory usage and inference latency by mapping model weights and activations to low-bit representations, making it an important technique for deploying VLAs on resource-constrained platforms. However, due to the characteristics of continuous closed-loop control, heterogeneous multimodal representations, and the sensitivity of action outputs to numerical precision, directly applying LLM-oriented quantization methods to VLAs may lead to substantial performance degradation.

For post-training quantization, QuantVLA [100] achieves training-free quantization through multi-

scale calibration. DA-PTQ [101] combines cross-space representation compensation with mixed-precision allocation to mitigate kinematic drift caused by low-bit quantization. QVLA [102] performs channel-wise quantization based on action-space sensitivity and jointly optimizes quantization and pruning. LiteVLA-Edge [103] further integrates low-bit quantization with a lightweight inference framework, enabling real-time VLA deployment on Jetson Orin with ROS2. These methods mainly rely on control-aware calibration and precision allocation to reduce quantization errors without retraining.

At the extreme end of the compression spectrum, HB-VLA [104] and BitVLA [86] push quantization toward very low bit-widths through two complementary routes: the former adopts policy-aware post-training quantization to preserve control performance under aggressive compression, while the latter employs a native ternary backbone with quantization distillation to reduce model storage.

Rather than fixing precision statically, a third line of work adapts it during training or inference. DyQ-VLA [105] dynamically adjusts quantization precision according to state sensitivity. SQIL [106] improves low-bit control accuracy by assigning greater training weight to critical states, while QAIL [107] enhances robustness to quantization error through quantization-aware fine-tuning. Together, these methods mitigate the impact of quantization error on action decisions via dynamic precision allocation and training-time robustness optimization.

3) *Distillation*: Distillation transfers knowledge from a large teacher model to a lightweight student model, enabling model compression while preserving multimodal understanding and action-generation capabilities. Compared with pruning and quantization, distillation can transfer semantic, structural, and action-related knowledge simultaneously, making it particularly suitable for capability transfer in embodied tasks.

ActDistill [108] models the action-prediction process through graph structures and introduces dynamic routing for action-aware computation-path selection, while VITA-VLA [109] transfers the execution capability of a small action expert to a vision-language model through action-space mapping and two-stage fine-tuning. Both approaches rely primarily on action-related supervision to preserve task understanding and execution capability after compression.

A second line of work targets the network structure itself. Shallow- π [110] reduces the Transformer depth of both the vision-language backbone and the Flow Matching action head. DySL-VLA [111] learns to dynamically skip redundant layers through distillation. AC²-VLA [98] combines cognitive reuse, token pruning, and selective layer execution with action-guided self-distillation. DualVLA [112] employs dual-teacher distillation to balance reasoning generalization and action accuracy, while HY-Embodied-0.5 [113] uses online policy distillation to transfer the capability of a large model to an edge-oriented variant.

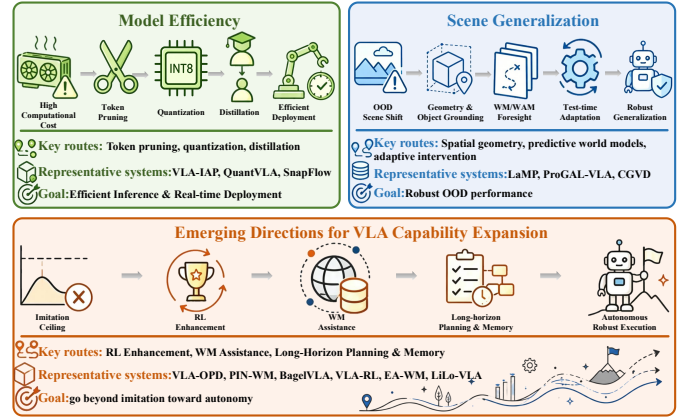


Fig. 4. Taxonomic overview of technical routes for VLA efficiency, scene generalization, and capability expansion. The figure summarizes three major research directions: improving model efficiency through token pruning, quantization, and knowledge distillation; enhancing scene generalization through spatial geometry, predictive world models, and adaptive intervention; and expanding VLA capabilities through reinforcement learning, world-model assistance, and long-horizon planning and memory. Key technical routes, representative systems, and target goals are summarized for each direction.

B. Scene Generalization

To address the performance degradation of VLAs in out-of-distribution environments, existing research has mainly formed three technical routes: spatial-geometric perception reduces a model’s reliance on surface features such as background by strengthening its understanding of object position and spatial relationships; world models improve a policy’s anticipatory capability by predicting environmental change and future states; and inference-time intervention combined with multi-skill composition enhances adaptation to new scenes and complex tasks without requiring model retraining.

1) *Spatial-Geometric Perception and Object-Centric Representation*: Traditional vision encoders are mainly optimized for semantic invariance and therefore have limited sensitivity to fine-grained spatial displacement and geometric structure. Recent studies address this limitation through three main directions: geometric-prior injection, multi-layer and multi-view alignment, and object-centric grounding.

For geometric-prior injection, LaMP [115] uses 3D scene flow as an action-generation condition to provide motion priors for action decoding. Robot-DIFT [116] distills geometric knowledge from diffusion models into the vision encoder, while GLaD [117] further extends geometric alignment to LLM hidden states to improve spatial reasoning. These methods enhance spatial modeling by introducing geometric information at either the action or encoder level.

A second direction leverages multi-layer and multi-view alignment. ROCKET [118] fuses spatial information from 3D vision foundation models through multi-layer feature alignment. PEAfowl [119] builds geometrically consistent cross-view representations and uses depth distillation to improve spatial perception in bimanual manipulation. PokeVLA [120] introduces geometry-aware tasks during

pretraining, achieving strong spatial generalization with a relatively small model. These methods exploit multi-layer and multi-view information to improve spatial representation.

Object-centric grounding addresses a related failure mode: VLAs often struggle to stably bind language instructions to scene entities under distractors or ambiguous instructions. ProGAL-VLA [121] enhances language-entity alignment through a 3D object-centric graph, while TAG [122] highlights target-relevant information and suppresses distractors through goal-erased comparison.

2) *World-Model-Driven Anticipatory Prediction*: Traditional VLAs are essentially reactive controllers that establish a static mapping directly from observation to action, with a decision process that lacks active modeling of the environment’s physical dynamic evolution. Introducing a World Model (WM) is a core idea for overcoming this limitation: by learning spatiotemporal physical priors from massive video data, a policy can gain, at decision time, an expectation of “how an action will change the state of the world.” Recent research has further evolved into WAMs, deeply integrating physical prediction with action generation. Around how the world model is introduced and how it is used at inference time, two current routes exist.

World-model-based methods. This route does not explicitly generate future visual observations, but instead injects video-prediction capability into a VLA’s encoding process in representational form. JEPA-VLA [123] and VLA-JEPA [124] are both based on the JEPA framework, differing mainly in how they fuse information: the former injects pretrained video-prediction embeddings into an existing VLA architecture via adaptive fusion; the latter adopts a strict leakage-free latent-state prediction mechanism, where the student path receives only the current observation rather than future frames, thereby forcing the model to learn genuine dynamic abstraction in latent space. AtomVLA [125] applies a predictive WM at the post-training stage, using a large language model to decompose high-level demonstrations into fine-grained atomic subtasks, and using a pretrained WM to score candidate action chunks in latent space. Its core contribution lies in mitigating long-horizon error accumulation through task decomposition and latent-space verification, forming an orthogonal complement to the aforementioned work targeting visual-scene perturbation.

WAM-based methods. WAM incorporates video tokens and action tokens into a unified generative framework, allowing the policy to decode actions synchronously while predicting future world-state evolution. OA-WAM [52] adopts an object-centric representation that decouples object identity from dynamic state, mitigating semantic confusion in global image prediction; HarmoWAM [50] combines the two strategies of “imagine before acting” and joint modeling, coordinating generalization ability and control precision through adaptive gating.

3) *Inference-Time Adaptive Intervention and Multi-Skill Compositional Generalization*: While the preceding approaches mainly improve generalization during training,

this line of work focuses on inference-time adaptation with frozen pretrained policies through perceptual intervention, online memory, and skill orchestration.

For perceptual and execution-level intervention, CGVD [126] reduces semantic distractions through target-distractor separation and image inpainting, whereas AFI [127] combines 3D affordances with proprioceptive feedback to detect execution stagnation and redirect the robot toward high-affordance regions.

At the memory and recovery levels, RoboHarness [128] integrates dual-memory RAG with an LLM orchestrator to retrieve behavioral priors and perform modular interventions without fine-tuning. SPR [129] instead converts language instructions into spatial subgoals and rewinds to recoverable states when execution stalls.

Beyond failure recovery, Act-Think-Abstain [130] dynamically switches among direct execution, further reasoning, and abstention according to physical and semantic anomalies. MergeVLA [131] enables multi-skill integration through mergeable task structures, while StarVLA- α [132] achieves strong multi-task generalization with a relatively simple unified architecture.

C. Capability Extension and Emerging Research Directions for VLAs

Although lightweight design and scene generalization lay the foundation for efficient VLA deployment, relying solely on the imitation-learning paradigm still has fundamental limitations for enabling robots to achieve autonomous, robust, long-term operation in the real world: a policy cannot exceed the distributional boundary of its demonstration data, lacks explicit reasoning about future states, and is prone to error accumulation and instruction drift as the task’s time horizon increases. To address this, recent research has built capability-enhancement frameworks along three complementary dimensions: reinforcement learning (RL) breaks through the performance ceiling of imitation learning via interaction-driven optimization; WM supports future-state prediction and policy improvement via internal modeling of environmental dynamics; and long-horizon reasoning improves cross-stage execution consistency by introducing explicit planning structures and memory mechanisms. Notably, these three are deeply interconnected: world models provide low-cost simulated environments for RL, RL in turn drives policy exploration within world models, and together the two provide reasoning support for long-horizon tasks.

1) *Reinforcement-Learning-Based VLAs*: Although pre-training and supervised fine-tuning have significantly improved VLA performance, imitation learning remains constrained by the quality and coverage of demonstration data, making it prone to error accumulation and insufficient adaptation in out-of-distribution scenarios. Reinforcement learning has therefore emerged as an important direction for VLA post-training, enabling further policy improvement through interaction-driven optimization. Existing studies mainly focus on optimization objectives,

exploration efficiency, data generation, task specialization, and system scalability.

VLA optimization has been improved through several forms of process-level supervision. VLA-OPD [134] applies online policy distillation with teacher-guided trajectory supervision. ARFM [135] introduces offline RL objectives for Flow Matching VLAs, while VLA-RL [136] and SRPO [137] extend rewards beyond terminal signals through process-level feedback.

Exploration efficiency is another important focus. IG-RFT [138] adapts exploration intensity based on interaction states, π_{RL} [139] models Flow Matching denoising as an MDP, and FASTER [140] filters candidate actions during denoising.

Recent studies have also explored RL-based data generation. DLR [141] improves trajectory diversity, while PLD [142] discovers failure cases through residual RL and distills recovery experiences into general-purpose VLAs.

Beyond optimization and data generation, several works target generalization and task specialization. SimpleVLA-RL [143] enhances generalization through exploration, GR-RL [144] specializes VLAs for precise manipulation, and RPD [145] and LongNav-R1 [146] improve policy adaptation and long-horizon credit assignment.

Reinforcement learning is evolving from a standalone post-training optimization technique into a broader framework that encompasses policy optimization, data generation, task specialization, and system scaling.

2) *World-Model-Based VLAs*: Many end-to-end VLAs rely primarily on current observations or short observation histories for action prediction, lacking explicit modeling of environmental dynamics. As a result, they are prone to prediction errors and error accumulation in long-horizon tasks. World models address this limitation by constructing predictive and rollable internal representations of the environment, allowing VLAs to estimate future state changes before execution and use these predictions to support action selection and policy optimization. Existing research mainly focuses on environmental dynamics representation, policy correction, and supervision generation.

World models represent environmental dynamics through latent, physical, and structured modeling paradigms. LaDi-WM [148] and DDP-WM [149] improve dynamics prediction through latent feature modeling and agent-aware decomposition, respectively. ContactGaussian-WM [150] and PIN-WM [151] incorporate contact modeling and differentiable physics to enhance physical prediction and Sim2Real transfer. H-WM [152], EA-WM [153], and OSVI-WM [154] further introduce structured intermediate representations for long-horizon execution and task generalization.

World models are also used for policy improvement through imagined rollouts and supervision generation. Hi-WM [155] generates recovery trajectories via imagined interaction and correction, while WM-Dagger [156] automates corrective data synthesis. WoVR [157] integrates world models with reinforcement learning through action-

conditioned prediction and joint policy optimization, extending their role from prediction to policy learning.

World models are evolving from future-state prediction modules into integrated mechanisms that support environment modeling, policy correction, supervision generation, and interactive optimization, providing stronger dynamic reasoning capabilities for long-horizon VLA decision-making.

3) *VLAs for Long-Horizon Tasks*: Long-horizon failures are usually not caused by a single bottleneck, but are jointly amplified by multiple types of deficiency within the execution loop: task goals are difficult to ground consistently across stages; key historical states are lost within a Markovian observation window; local errors accumulate over time, leading to irreversible trajectory deviation; and there is a lack of reliable detection and recovery mechanisms after execution failure. Recent research addresses these root causes of failure through systematic exploration along four dimensions: hierarchical planning, persistent memory, adaptive recovery, and predictive unified reasoning.

Long-horizon execution is often addressed through hierarchical planning. VLAs-as-Tools [158] coordinates a high-level VLM with a specialized VLA for planning and execution. Anticipation-VLA [159] generates recursive subgoals, while LoHo-Manip [160] and LiLo-VLA [161] improve long-horizon control through progress prediction and hierarchical motion decomposition.

Persistent memory mechanisms improve long-term dependency handling. HELM [162] enables memory recovery through retrieval and rollback, while VQ-Memory [163], TempoFit [164], and KC-VLA [165] preserve historical information through compressed representations, temporal context extension, and keyframe retrieval.

Adaptive recovery methods enhance robustness after execution failures. Critic in the Loop [166] detects failures and triggers replanning, RoboClaw [167] introduces self-resetting recovery loops, and Goal2Skill [168] integrates planning, verification, and execution into a continuous recovery framework.

Some approaches further explore unified predictive reasoning. BagelVLA [169] jointly models planning, prediction, and action generation, while AtomVLA [125] combines subtask decomposition with world-model-based action scoring. These methods reduce information fragmentation but remain constrained by joint training and inference efficiency.

Long-horizon VLA research mainly improves execution consistency and robustness through hierarchical task decomposition, persistent memory, closed-loop recovery, and future-state prediction.

IV. Core Challenges Facing VLAs

VLAs have already made significant progress on short-horizon, standard tasks, but there remains a considerable gap before they become generalizable, deployable, continuously evolving general-purpose embodied intelligence systems. These gaps correspond to structural bottlenecks

along seven dimensions: model capability, deployment efficiency, data infrastructure, physical understanding, task reasoning, system architecture, and data sourcing. These seven categories of challenges cover the main limitations current VLAs face in terms of efficiency, generalization, reasoning, interaction, and deployment, together reflecting the key problems that still need to be solved on the path from experimental settings to real-world application.

A. Limitations in Cross-Scene Generalization

VLAs perform stably within the training distribution but degrade markedly when facing unseen tasks, scenes, embodiments, or long-tail perturbations. The underlying cause is that models tend to learn appearance statistics and trajectory regularities embedded in the training data rather than transferable knowledge of object relations, spatial structure, and action. Vision encoders typically fuse target entities, background texture, and camera viewpoint into a single implicit representation, so action prediction remains sensitive to appearance change; when object position, scene layout, or embodiment shifts, the correspondence between instruction, visual target, and action destabilizes accordingly. Under instruction rephrasing, target distraction, or ambiguous expression, models still tend to fall back on visual cues and trajectory priors learned during training rather than fully grounding their predictions in language. The generalization bottleneck is therefore not purely a matter of visual encoding capacity, but reflects deeper deficits in object-relation modeling, spatial-language grounding, and cross-embodiment action alignment.

B. Fragility in Long-Horizon Task Execution

Long-horizon manipulation is difficult not simply because context length is limited, but because most models lack a verifiable representation of task progress and a mechanism for closed-loop recovery. VLAs are largely reactive mappings from observation to action, with no capacity to self-evaluate execution state; a failure at one node can therefore propagate irreversibly through the rest of the sequence. Task progress is encoded as a continuous action stream rather than a discrete state, leaving the model unable to judge whether a subtask has completed or to retrieve relevant historical states for downstream decisions. Subtask decomposition and memory augmentation offer partial mitigation, but they are typically bolted on as external, offline modules rather than integrated into a stable closed-loop mechanism within the end-to-end backbone.

C. Bottlenecks in Lightweight Deployment and Real-Time Control

Because VLAs are built on large-scale vision-language backbones, the semantic understanding they gain comes at the cost of inference latency, which conflicts with the real-time and edge-deployment demands of closed-loop robotic control. Compression methods tailored to

VLAs are beginning to appear, but achieving extreme compression without sacrificing action-control precision remains a substantial engineering problem. Few methods explicitly address control frequency or closed-loop stability for action prediction. Hierarchical architectures that pair slow, high-level planning with fast, low-level execution are still immature, and task-adaptive computation and asynchronous inference remain exploratory rather than systematized solutions to the performance-efficiency trade-off.

D. Lack of Continual Learning Ability under the Static-Data Paradigm

Most VLAs are trained offline on static datasets and deployed without any mechanism for absorbing new tasks, failure cases, or recovery experience encountered after deployment. Models cannot learn from their own deployment-time errors, nor can they adapt promptly to the new scenes and failure modes that open environments continually produce. Online interaction, failure-trajectory generation, and policy retraining have each been explored to address this, but a stable pipeline that moves from deployment to feedback collection, filtering, and retraining has yet to be established at scale.

E. Insufficient Modeling of Physical Laws

VLAs rely predominantly on data-driven statistical learning and lack explicit representations of contact dynamics, support relations, friction, and mass distribution. Many manipulation failures trace not to perception or semantic errors but to the model's inability to predict the physical consequences of an action. This limitation is most evident in fine-grained tasks such as grasp stability, assembly alignment, and deformable-object manipulation. Incorporating physical priors and affordance knowledge into representation learning remains an open problem, and its absence limits both the stability and the interpretability of current models in contact-rich settings.

F. Lack of Embodied Agentic Capability

Most VLAs remain, at core, a policy mapping from vision-language input to action; agentic capabilities such as task decomposition, long-term memory, tool invocation, and active exploration are added externally rather than built in. No stable, unified architecture yet coordinates high-level reasoning models, low-level VLA policies, and world models for multi-stage planning and human-robot collaboration. The interfaces between high- and low-level information, cross-module memory sharing, and safety constraints all remain unresolved. Until these are addressed, robots will struggle to move beyond executing a single policy toward closed-loop autonomy with planning, verification, and recovery.

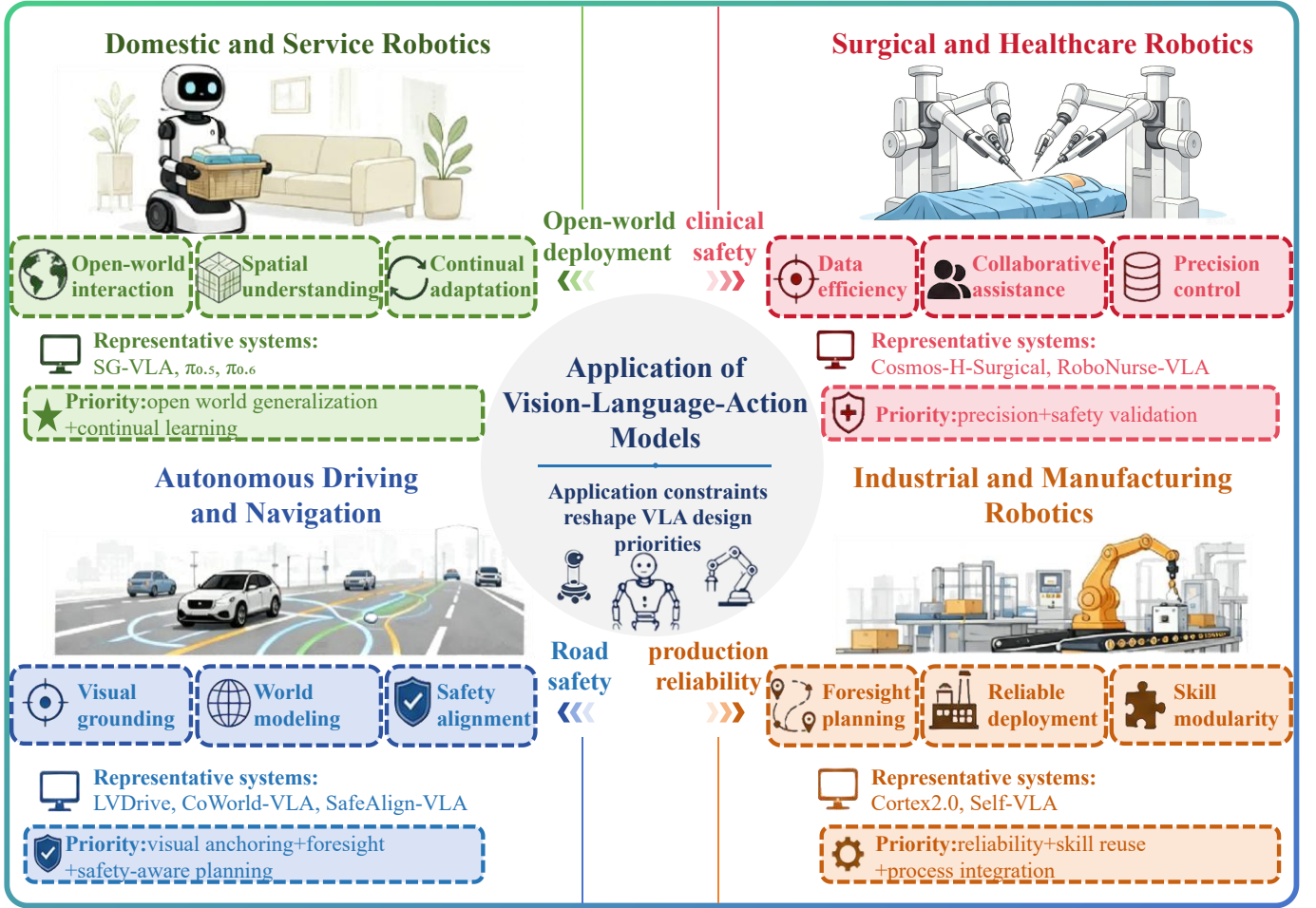


Fig. 5. Application landscape and design priorities of VLAs across four representative domains. The figure illustrates the domain-specific requirements and corresponding VLA design priorities in domestic and service robotics, surgical and healthcare robotics, autonomous driving and navigation, and industrial and manufacturing robotics, together with representative systems in each domain.

G. Scarcity of Real-World Data and Insufficient Credibility of Alternative Data Sources

Robotic manipulation data depends on real hardware and expert demonstration, which makes collection expensive and hard to scale; differences in kinematic structure and action space across embodiments further limit reuse across platforms, and coverage is thin for complex contact manipulation and deformable-object handling in particular. Human video and simulation offer cheaper alternatives, but each has its own limitation: human video requires bridging a cross-embodiment gap between human and robot kinematics, while simulation is bounded by the sim-to-real gap in both appearance and dynamics. More fundamentally, neither source provides a reliable supervisory signal linking visual dynamics to executable robot action, which constrains how directly either can be put to use.

V. Application Scenarios for VLAs

As illustrated in Fig. 5, the evolution of VLA capabilities must ultimately be tested through deployment in the real physical world. Different application scenarios impose

different core requirements on robots: household service emphasizes continuous adaptation in an open world; surgical robots emphasize predictable behavior under extreme precision constraints; autonomous driving emphasizes visual grounding and anticipatory decision-making; and industrial manufacturing emphasizes reliability and flexibility under large-scale deployment. This section focuses on these four representative scenarios, illustrating how the technical evolution of VLAs responds to the real-world constraints of different domains.

A. Household and Service Robots

The core challenge of the household environment lies in its openness: object types are diverse, spatial layouts vary, and task requirements are highly individualized, requiring the robot to accomplish spatial understanding, instruction execution, and long-term autonomous adaptation without structured priors. This requires VLAs to possess not only fine-grained local manipulation ability, but also stable performance in unseen environments, along with the ability to continuously improve as deployment time progresses. Around this need, recent research unfolds along three

levels: at the level of spatial representation, SG-VLA [170] enhances the robot’s global-local perception of household scenes by fusing multi-view observations with auxiliary supervisory signals; at the level of open-world generalization, $\pi_{0.5}$ [171] enables the model to transfer knowledge from other environments to unseen household layouts and objects through joint training on heterogeneous, cross-scene data; at the level of long-term deployment, $\pi_{0.6}^*$ [172] continuously integrates online interaction experience and expert corrective feedback, driving robots to improve execution efficiency on complex, long-horizon tasks through real-world use. These studies indicate that household service applications are gradually shifting VLA systems from static deployment after offline training toward continual adaptation through long-term interaction.

B. Surgical and Medical Assistance Robots

Surgical and medical assistance scenarios impose requirements on precision, safety, and data availability that exceed those of most application domains: operational error must be controlled at the sub-millimeter scale, behavior must be highly predictable, and the privacy sensitivity of patient data limits the scale of available training data. This combination of constraints determines that the technical route for medical VLAs differs from general-purpose models that pursue broad generalization, tending instead toward specialized structural design and data-efficient learning. In this direction, researchers have used lightweight, specialized network structures to demonstrate that models can achieve stable, fine-grained manipulation with very little demonstration data [173]; Cosmos-H-Surgical [174] constructs a surgery-specific world model combined with inverse-dynamics inference, converting large volumes of unlabeled surgical video into usable training signal, alleviating the scarcity of action-labeled data; RoboNurse-VLA [175] integrates visual segmentation with language understanding to achieve surgical-instrument recognition and real-time delivery based on natural spoken instructions, providing a new interaction mode for human-robot collaboration in the operating room. The development path of medical VLAs reflects a domain characteristic of being “specialized rather than general, precise rather than broad.”

C. Autonomous Driving and Navigation

Autonomous driving requires a model to complete scene understanding, risk assessment, and trajectory decision-making in real time under high-dimensional perceptual input, where errors can lead to irreversible safety consequences; this domain therefore imposes comprehensive requirements on the use of visual information, long-horizon scene prediction, and risk-sensitive decision-making mechanisms. Around these requirements, research on driving VLAs has formed several technical threads: some work aims to correct the model’s over-reliance on language and vehicle state, redesigning the form of action supervision and the trajectory-decoding interface to encourage the

model to make fuller use of visual information for planning [176], [177]; other work introduces world-modeling capability, enhancing the model’s anticipatory ability for complex traffic scenes through multi-expert scene representations or generative future-frame prediction, alleviating the lack of foresight in reactive policies [178], [179]; for safety enhancement, related studies introduce risk identification, behavior correction, and external safety priors into the planning process by constructing counterfactual negative samples, generating and evaluating candidate trajectories for subsequent optimization, and retrieving historical driving experience. These mechanisms improve the decision safety and robustness of driving-oriented VLAs in complex scenarios and long-tail conditions [180]–[182]; in addition, a batch of work explores unified architectures for language reasoning and continuous control, using low-entropy meta-actions, single-forward-pass joint generation, or mixture-of-experts parameter decoupling to balance performance with inference efficiency and multi-task capability [183]–[185]. Overall, recent driving-oriented VLA research has placed increasing emphasis on adapting model architectures to the specific requirements of perception, reasoning, and control in autonomous driving.

D. Industrial and Manufacturing Robots

The core requirement that industrial manufacturing scenarios place on VLAs is reliability and flexibility under conditions of large-scale deployment: robots must maintain stable operation in unstructured environments with frequent clutter and occlusion and dense contact, while also adapting to constantly changing task requirements at low configuration cost. Around these goals, existing research mainly unfolds along two complementary paths. The first line of work focuses on improving execution reliability in complex tasks, generating and evaluating candidate trajectories in a visual latent space and selecting subsequent actions based on expected success rate and execution efficiency, thereby mitigating error accumulation and failure propagation in long-horizon tasks under reactive policies. Meanwhile, studies targeting real production lines form a deployment-driven iterative optimization loop through continuous on-site data collection, sample screening, policy fine-tuning, performance evaluation, and recovery-data supplementation. This enables industrial VLAs to move beyond one-time training followed by direct execution toward deployment strategies that combine anticipatory planning with continual optimization [186], [187]. The second line of work focuses on enhancing task configuration and skill-reuse capability, decomposing complex assembly or disassembly processes into callable, specialized skills through structured skill libraries and language-driven task scheduling, dynamically switching based on task context, thereby improving the robot’s adaptability to high-precision, heavily contact-rich, and multi-stage operations [188], [189]. Overall, industrial VLAs are not blindly pursuing a fully end-to-end unified policy, but are gradually adopting a compositional

architecture in which a foundation model handles task understanding and planning while specialized skill modules perform precise operations. This development direction better matches the long-term engineering demands of industrial systems for reliability, maintainability, and task flexibility.

VI. Future Outlook

The main limitation of current VLAs is no longer whether they can generate plausible actions under familiar conditions, but whether they can remain reliable as tasks, environments, and interaction histories evolve. Long-horizon execution, continual adaptation, and experience reuse expose limitations that cannot be addressed by model scaling alone. Future progress will therefore depend on coordinated advances in model architecture, data, physical modeling, deployment, and agentic learning. Based on the preceding analysis, we discuss seven directions that may shape the next stage of VLA development as shown in Fig. 6.

A. Generalization: From In-Distribution Success Toward Robustness in Open Environments

Generalization ability is the core prerequisite for VLAs to become general-purpose robots. Existing models perform well within the training distribution, but performance still drops markedly when facing unseen tasks, scenes, robot embodiments, and long-tail perturbations. Future research therefore needs to focus on cross-task, cross-scene, cross-embodiment, and long-tail robustness. Cross-task generalization requires models to learn reusable skills and compose them; cross-scene generalization requires models to adapt to different environments such as households and factories; cross-embodiment generalization requires models to adapt to different robot platforms. Long-tail robustness concerns real-world scenarios such as occlusion, abnormal layout, human interference, and failure recovery. Future work needs to develop unified action representations and transferable control interfaces, build a data ecosystem that includes both success and failure cases, and design more challenging compositional-perturbation evaluations. In addition, by modeling environmental dynamics and action consequences, WAM is expected to improve a model's prediction and transfer ability in unseen environments. Ultimately, the goal of VLAs will shift from high success rates on fixed benchmarks toward stable, reliable behavior in open environments.

B. Lightweight Deployment: From Large-Scale Models Toward Real-Time, Deployable Embodied Policies

Current VLAs often rely on large-scale vision-language models; while this improves semantic understanding, it also brings high computational cost and inference latency. Robotic control must meet real-time and edge-deployment requirements, making lightweight design a key direction. Methods such as low-rank adaptation, quantization, pruning, and knowledge distillation can reduce training and

deployment cost. At the same time, mechanisms such as action-chunk prediction, asynchronous inference, and streaming control also help improve closed-loop control efficiency. Future VLA systems may increasingly adopt hierarchical architectures that separate computationally intensive reasoning from time-critical control. A high-level model can perform task interpretation and planning at a lower update frequency, while a lightweight low-level policy executes actions at the rate required for closed-loop control. Such a division of computation could reduce the cost of invoking large models at every control step while retaining their reasoning capability when needed, providing a practical trade-off between model capacity and real-time execution.

C. Data Flywheel: From Static Datasets Toward a Continuously Evolving Data Engine

Existing VLAs mainly rely on static datasets for training, but the real environment continuously produces new tasks and failure cases, and fixed data cannot support continual evolution. Future work therefore needs to build a data flywheel to enable continual learning. A data flywheel updates the model by continuously recycling deployment experience: as a robot executes tasks, it records success and failure cases, which are filtered and used in the next round of training, after which the improved policy is redeployed in practice. In the future, competition among VLAs will be not only competition among models, but also competition in data infrastructure. Combined with WAM and world models, it becomes possible to predict failure risk and filter out implausible trajectories, thereby improving data-utilization efficiency and safety, evolving VLAs from one-off trained models into continuously evolving embodied learning systems.

D. Physical Knowledge Enhancement: Toward More Physically Grounded Policy Learning

Data-driven VLA policies can capture useful correlations from large-scale demonstrations, but such representations do not necessarily encode the physical constraints that govern robot-environment interaction. This limitation becomes particularly relevant in contact-rich manipulation, assembly, and deformable-object tasks, where successful execution depends on factors such as support, friction, mass distribution, and contact dynamics.

Future research may therefore benefit from incorporating physical structure more explicitly into policy learning. Possible directions include introducing physical priors and affordance information into learned representations, using world models to predict action-conditioned state transitions, and exploiting physics-based simulation to provide additional supervision for interactions that are difficult to collect at scale in the real world. Such approaches could improve the consistency between predicted actions and the physical evolution of the environment.

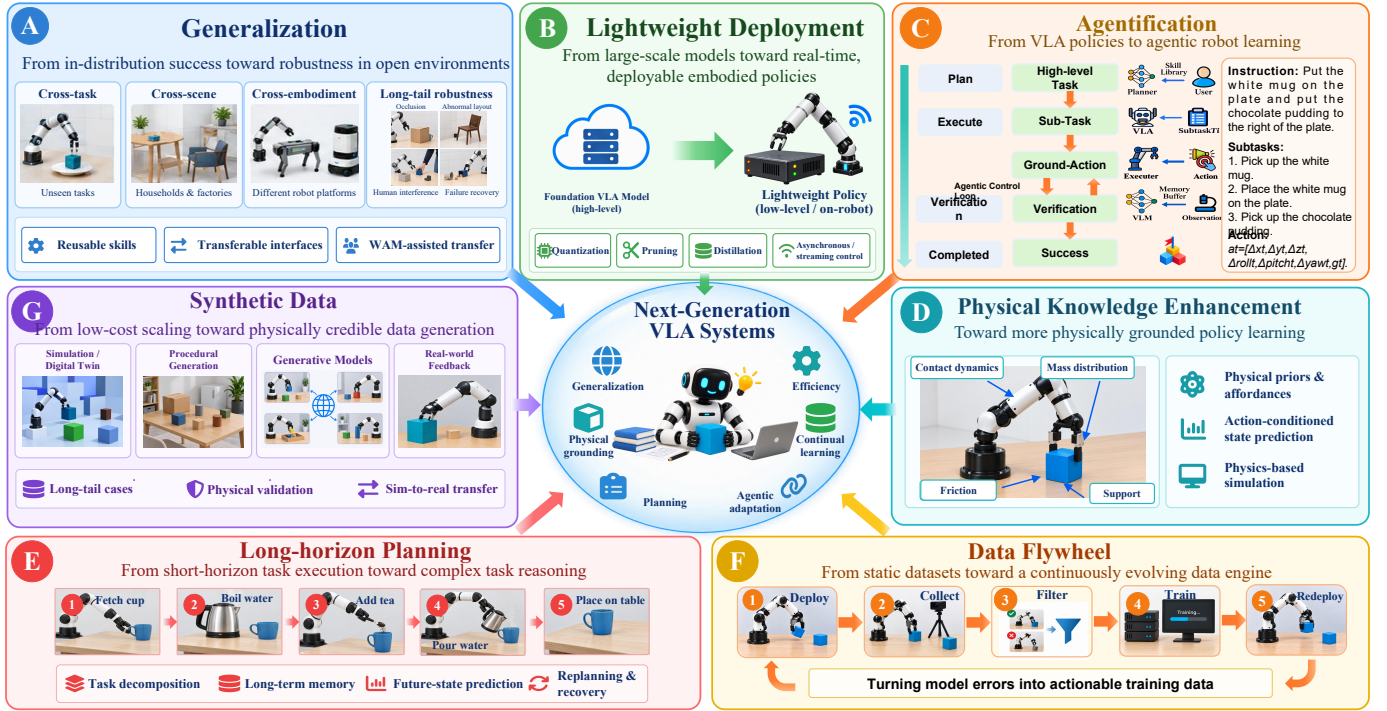


Fig. 6. Open research directions and critical challenges in VLAs. A systematic summary of seven foundational pathways enabling next-generation generalist embodied models: open-world generalization, lightweight deployment and real-time control, continuous data flywheels, physical knowledge enhancement, long-horizon planning and memory, autonomous embodied agentification, and physically faithful synthetic data generation.

E. Long-Horizon Planning: From Short-Horizon Task Execution Toward Complex Task Reasoning

Current VLAs perform well on short-horizon manipulation tasks, but real-world tasks often involve multi-stage decision-making and long-term goals. Household service and industrial assembly tasks, for example, require task decomposition, state memory, and error recovery. Future VLAs need mechanisms for decomposing extended tasks, tracking execution progress, and revising plans as intermediate outcomes unfold. World models can support this process by predicting future states and evaluating the consequences of candidate actions. Long-horizon tasks also remain vulnerable to error accumulation and state forgetting, motivating the use of long-term memory, closed-loop replanning, and post-execution reflection. Evaluation protocols should likewise move beyond a single success rate to consider process quality, recovery ability, and long-term consistency.

F. Agentification: From VLA Policies to Agentic Robot Learning

Future robotic systems may increasingly embed VLA policies within Agentic Robot Learning frameworks rather than treating the policy itself as the entire decision-making system. In this setting, the VLA provides task-conditioned action generation, while planning, memory, execution monitoring, and feedback-driven adaptation operate around the policy to support closed-loop behavior over extended interactions.

An embodied agent harness can provide the system-level structure needed to coordinate these components. Rather than replacing the VLA, the harness manages the interaction among the policy, memory, callable skills, world models, and execution feedback, and can trigger replanning or recovery when task execution deviates from the intended outcome. This separation allows agentic capabilities to be organized at the runtime level without requiring every function to be encoded within a single policy model.

From a learning perspective, successful and failed executions can be retained as persistent experience and reused for later planning, recovery, or policy adaptation. Agentic Robot Learning can therefore be viewed as extending VLA-based control from action generation toward continual closed-loop improvement, while the harness provides the execution infrastructure through which such adaptation can be maintained during long-term deployment.

G. Synthetic Data: From Low-Cost Scaling Toward Physically Credible Data Generation

Robot data collection is costly, especially for complex manipulation and hazardous scenarios, making synthetic data an important avenue for scaling VLA capability. Through simulated environments, procedural generation, and generative models, large-scale training data can be obtained at low cost. The value of synthetic data lies not only in scale, but also in supplementing long-tail scenarios that are underrepresented in real data, such as abnormal layouts, failure recovery, and hazardous tasks. It can

also support tasks such as cross-embodiment transfer and complex contact manipulation. However, discrepancies between simulation and reality may cause a model to learn incorrect regularities. Future research therefore needs to focus on data quality, physical credibility, and real-world transferability, calibrating through physical verification and real feedback. Future work may explore integrating synthetic data generation with deployment feedback and world models, so that failure cases observed in practice can guide the generation of targeted training samples.

Taken together, these directions suggest that future VLA development will increasingly extend beyond the optimization of individual policy models. Progress will also depend on how models interact with evolving data, physical knowledge, predictive representations, and accumulated experience during deployment. Improvements in generalization and efficiency may broaden the range of environments in which VLAs can operate, while long-horizon reasoning, continual adaptation, and agentic learning could enable robots to maintain and revise their behavior over extended interactions. Synthetic data and world modeling may further complement real-world experience by providing additional supervision for difficult or sparsely observed situations. These developments may gradually move VLAs toward more adaptive and reliable embodied systems capable of operating beyond fixed tasks and benchmark conditions.

VII. Conclusion

The studies reviewed in this survey show that VLAs have benefited substantially from advances in large-scale vision-language pretraining, action-generation mechanisms, and robotic data. These developments have broadened the range of tasks and environments that can be addressed within a unified policy framework. However, benchmark performance alone provides only a partial indication of real-world capability, since reliable deployment also depends on closed-loop stability, robustness to distribution shifts, and computational efficiency.

Our analysis further suggests that no single action-generation paradigm is uniformly preferable across robotic tasks. Autoregressive, diffusion-based, and direct-regression approaches involve different trade-offs in representation, control precision, and inference cost, and their suitability depends on the requirements of the target application. Likewise, improvements in model efficiency or data coverage alone do not guarantee reliable behavior beyond the training distribution.

The emergence of VAMs and WAMs reflects a broader shift toward incorporating temporal evolution and predicted future states into robot policy learning. Their significance lies in using predictive information to support action generation and decision-making rather than simply adding separate prediction modules. Overall, further progress toward general-purpose embodied intelligence will require closer coordination among action generation, predictive modeling, and closed-loop interaction with the physical world.

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